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Highly Sensitive N-FK51A/Cu/ZnSe/BP-Based SPR Biosensor for Dengue Virus Detection: A TMM and FEM Approach

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***Abstract*—This work demonstrates the design and optimization of a multilayer prism-based Surface Plasmon Resonance (SPR) biosensor intended for the early and accurate detection of blood cells infected with the dengue virus. Our proposed structure utilizes an N-FK51A prism that is coated with copper (Cu), zinc selenide (ZnSe), and a monolayer of black phosphorus (BP). This configuration creates an optimized plasmonic interface that improves light-matter interaction. To determine the optimal thickness of each layer, we applied Finite Element Method (FEM) simulation in COMSOL Multiphysics and Transfer Matrix Method (TMM) analysis in MATLAB. The biosensor shows a peak angular sensitivity of 327.71°/RIU, full width at half maximum (FWHM) of 1.89 deg., 0.7874 deg^{-1} detection accuracy (DA), and a quality factor (QF) of 156.68 RIU^{-1} for dengue-affected hemoglobin cells. Close agreement of results obtained from FEM and TMM confirms the computational approaches accuracy and the reliability of the proposed model. Overall, the proposed N-FK51A/Cu/ZnSe/BP biosensor offers a cost-effective and promising platform for rapid detection dengue virus.**

Keywords—Surface Plasmon Resonance (SPR), plasmonic resonance, biosensor optimization, transfer matrix method (TMM), finite-element method (FEM), quality factor (QF), detection accuracy (DA).

I. Introduction

Transmitted through the Aedes aegypti mosquito, dengue is the most common arthropod-borne viral disease which can be found in tropical and subtropical regions around the world. The dengue virus is estimated to infect about 390 million people each year. Of these, nearly 96 million cases progress to severe forms of the disease, such as dengue hemorrhagic fever (DHF) or dengue shock syndrome (DSS), according to reports from the World Health Organization (WHO) [1]. As specific antiviral treatments are absent, accurate and early diagnosis of dengue is essential to prevent complications and lower death rates. Unfortunately, conventional diagnostic methods such as enzyme-linked immunosorbent assay (ELISA), reverse transcription polymerase chain reaction (RT-PCR), and lateral flow immunoassays (LFIA) were found to be ineffective. These techniques are generally slow, expensive, and require a lot of labor. Furthermore, they are less effective at detecting early-stage infections [2]. These shortcomings underscore the necessity for faster and accurate diagnosis approaches. Surface Plasmon Resonance (SPR) sensors are gaining attention for their potential in early disease detection. SPR operates by detecting and monitoring alterations in the refractive index at a metal-dielectric interface, which occur due to biomolecular binding. Unlike ELISA and PCR, SPR requires less sample preparation and no labeling of biomolecules [3].

The materials used in an SPR sensor design have a significant impact on its performance. Hybrid multilayer prism-type Kretschmann configuration sensors combining metals like Ag, Au and Cu along with dielectric oxides such as ZnO, ZnSe and $BaTiO_3$ and 2D materials such as Black Phosphorus (BP), Graphene (GO), and MoS_2 have achieved sensitivities in the range of 300 to 500°/RIU [4]. The BK7/Ag/TiO2/BP structure used in the SPR sensor described by S. Das et al. [5] offers a sensitivity of 309.742°/RIU, addressing fabrication complexity and operational stability limits. Meanwhile, N. P. Singh et al. [6] developed an SPR sensor that used perovskite and silicon nitride nanomaterials for detecting dengue cells, which attains a sensitivity of 300°/RIU. Yesudasu et al. [7] also reported a SPR sensor with a simple BK7/Ag/AlN/2D material structure, which reached an improved sensitivity of 310.29°/RIU for dengue virus detection.

Here, we have introduced a prism-based multilayer SPR biosensor designed for the early detection of dengue-infected blood cells. The novelty of this work lies in the integration of an inexpensive Cu/ZnSe/BP multilayer structure on an N-FK51A prism, which is systematically optimized and validated using both TMM and FEM approaches. The N-FK51A prism increases angular sensitivity and reduces optical loss, while the Cu coating allows for strong plasmonic resonance at a lower cost compared to gold [6,8]. The ZnSe layer ensures stable and efficient dielectric coupling [9], and interaction between the sensor surface and biomolecules is enhanced by using a BP layer, because of its large surface area [10]. This combination of materials improves light confinement and increases biomolecule adsorption, thereby improving the sensor's performance. Performance parameters, such as sensitivity, QF, DA, and FWHM of our proposed model have been evaluated at a 633nm wavelength. The findings show that the sensor is reliable for detecting various blood components associated with dengue.

II. Design Methodology

A. Architectural Modeling

The proposed SPR sensor for detecting the dengue virus comprises of five unique layers. Fig. 1(a) shows the structural design of the device. The multilayer structure consists of an N-FK51A prism coated with a 45 nm thick Cu film, followed

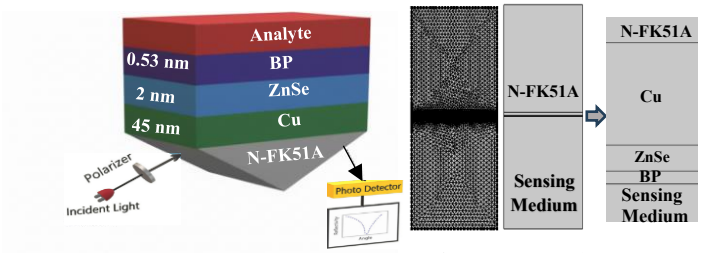
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by a ZnSe layer about 10nm thick. Finally, a thin BP layer with a thickness of 0.53nm is placed at the uppermost layer. Each material has been selected to best support the sensor's optimal performance in terms of sensitivity, efficiency, and compatibility with SPR detection methods. In the initial layer, light with a wavelength (λ) of 633 nm is coupled through an N-FK51A prism substrate. The following formula can be used to determine the RI of the N-FK51A prism [8].

$$n_{N-FK51A}(\lambda) = \sqrt{1+\frac{0.971247817\lambda^2}{\lambda^2-0.00472301995}+\frac{0.216901417\lambda^2}{\lambda^2-0.0153575612}+\frac{0.904651666\lambda^2}{\lambda^2-168:68133}} \quad (1)$$

TABLE I.
REFRACTIVE INDEX OF SELECTED MATERIALS AT 633nm

Material	Refractive Index (RI)	Ref.
N-FK51A	1.4853	[8]
Cu	0.0369+4.5393i	[6]
ZnSe	2.57+0.0001i	[9]
BP	3.5+0.01i	[10]



(a) (b) (c)

Fig. 1. (a) Schematic representation of the proposed N-FK51A/Cu/ZnSe/BP hybrid structure–based SPR biosensor for Dengue Detection (b) General mesh view of N-FK51A/Cu/ZnSe/BP structure (c) Simulated structural model of the proposed biosensor developed in COMSOL

Fig. 1(a) presents the structural design of the proposed SPR biosensor Fig. 1(b) presents the mesh size of finer profile of the proposed sensor. A simulated five-layer biosensor structure and its enlarged view are shown in Fig. 1(c).

The physical vapor deposition (PVD) technique can be used to deposit the Cu layer. The Drude model is used to calculate the dielectric constant of the 2nd layer of Cu, which can be expressed as [3]:

$$\varepsilon_{Cu}(\lambda) = 1 - \frac{\lambda^2\lambda c}{\lambda_p^2(\lambda_c+i\lambda)} \quad (2)$$

where the plasma wavelength, λ_p, is 1.3617×10^{-7}m, and the Cu collision wavelength, λ_c, is 4.0852×10^{-5}m.

The following Sellmeier equation is employed to determine the RI of ZnSe at 633 nm RI [9]:

$$n^2(\lambda) = A + \frac{B\lambda^2}{\lambda^2-C^2} \quad (3)$$

where λ is the wavelength in microns. For ZnSe, the best values of the parameters are A=4.00, B=1.90, and c^2=0.113.

Finally, because of its remarkable production capabilities and effectiveness at lower temperatures, the CVD method is used to deposit a layer of BP on top of the surface. Refractive index of BP [5]:

$$n_{BP}(\lambda) = A + \frac{B}{\lambda^2} + \frac{C}{\lambda^4} \quad (4)$$

A. Mathematical modeling

The proposed SPR sensor detects the dengue virus using a composite structure of N-FK51A/Cu/ZnSe/BP. A key operational feature of the sensor is the measurement of the reflectance intensity of TM-polarized (p-polarized) light. Therefore, the reflection coefficient for p-polarized light is represented by [11]:

$$R_p = |r_p^2| \quad (5)$$

The TMM method is used to compute the reflection coefficient (r_p) of a multilayer structure.

$$r_p = \frac{(L_{11}+L_{12}P_N)p_1-(L_{21}+L_{22}p_N)}{(L_{11}+L_{12}P_N)p_1+(L_{21}+L_{22}p_N)} \quad (6)$$

Since we are dealing with a multilayer coating, the transfer matrix formula, $A_{ij,}$ can be denoted as follows [12]:

$$A_{ij} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \left(\prod_{k=2}^{N-1} A_k\right) \quad (7)$$

where,

$$A_k = \begin{bmatrix} \cos U_k & \frac{-i\sin U_k}{p_k} \\ -iP_k \sin U_k & \cos U_k \end{bmatrix} \quad (8)$$

here, U_k and P_k presents the phase factor and the admittance corresponding to a specific phase, respectively.

B. Performance Parameters

To achieve optimal performance, the SPR sensor is evaluated based on several parameters. Calculations for angular sensitivity (S), FWHM, DA and QF are determined using the following equations [5]:

$$S = \frac{\Delta\theta_{SPR}}{\Delta n} \quad (9)$$

$$FWHM = \theta max - \theta min \quad (10)$$

$$DA = \frac{1}{FWHM} \quad (11)$$

$$QF = \frac{S}{FWHM} \quad (12)$$

III. RESULTS AND ANALYSIS

A. Detection of Dengue Virus in Human Blood

The TMM analysis was performed using MATLAB software to determine the resonance angle, reflectance, and corresponding angular sensitivity. To verify these findings, the sensor structure was further modeled in COMSOL Multiphysics (v6.2) using the FEM simulation framework. Table II shows the refractive index (RI) [13] and the resonance angles for healthy and dengue-infected blood cells. On top of the proposed sensor, samples of healthy platelets, plasma, hemoglobin, and dengue-infected blood cells were arranged. The sensor exhibited angular sensitivity of 153.84°/RIU for infected plasma, 235.4545°/RIU for infected platelets, and 327.71°/RIU for infected hemoglobin blood cells.

TABLE II.
REFRACTIVE INDEX OF NORMAL AND DENGUE-INFECTED BLOOD CELLS

Sensing Medium	Reference Wavelength	RI [13]	Resonant Angle(deg.)
N-PLT	633 nm	1.390	83.34
A-PLT		1.357	75.57
N-PLS		1.350	74.39
A-PLS		1.337	72.39
N-Hgb		1.365	77.04
A-Hgb		1.400	88.51

Table III summarizes the performance metrics for the infected blood components. According to Table III, the infected hemoglobin sample shows the highest angular sensitivity and QF, whereas the infected plasma sample demonstrated the highest DA and FWHM.

TABLE III.
PERFORMANCE METRICS OF THE PROPOSED SENSOR FOR INFECTED BLOOD COMPONENTS

Sensing Medium	Angular Sensitivity (deg./RIU)	DA ($deg.^{-1}$)	FWHM (deg.)	QF (RIU^{-1})
Affected platelet	235.45	0.45	1.06	145.13
Affected plasma	153.84	0.94	2.20	106.78
Affected hemoglobin	327.71	0.52	1.89	173.39

B. Analysis of Electric and Magnetic Field Distribution

The electromagnetic field distribution across the interfaces of the proposed SPR biosensors is examined using the FEM approach. Fig. 2(a) and Fig. 2(b) exhibit the instantaneous electric field (E_y) distribution at resonance angles of 77.04° and 88.51° for RI 1.365 and 1.400, respectively. The field intensity shows stronger confinement with higher RI and is focused close to the metal dielectric contact. The corresponding magnetic-field (H_z) distribution at the same resonance angles and refractive indices is presented in Fig. 3(a) and Fig. 3(b).

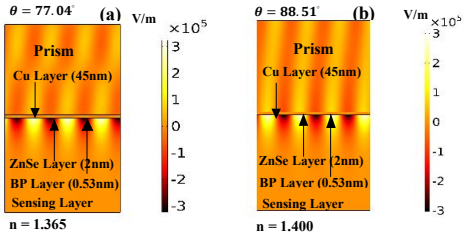


Fig. 2. Instantaneous Electric field (E_y) distribution at resonance angle (a) 77.04° and (b) 88.51° of RI 1.365 and 1.400.

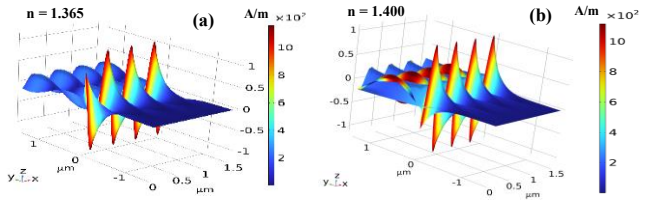


Fig. 3. A three-dimensional view of the magnetic field's (H_z) component at resonance angle (a) 77.04° and (b) 88.51° of RI 1.365 and 1.400.

Fig. 4 shows the electric field distribution, where the maximum electric field norm reaches 3.19 ×10^5 for the RI 1.400 at the resonance angles 88.51° and a non-resonance angle 80°, clearly indicating that the electric field strength is significantly lower at non-resonance angles.

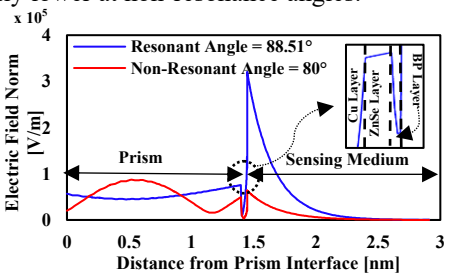


Fig. 4. Electric field distribution curve for Affected Hgb at RI 1.400 for the resonant angle 88.51° and non-resonant angle 80°.

C. Effect of variation in Cu Layer Thickness

The thickness of each layer influences the resonance conditions, resulting in variations in reflected light intensity and thereby affecting the device's sensitivity. In this work, we examine how different thicknesses of the Cu layer affect the performance of an SPR sensor, while ensuring all other layers maintain a uniform thickness. Table IV shows that a 45 nm thick Cu layer provides the highest sensitivity.

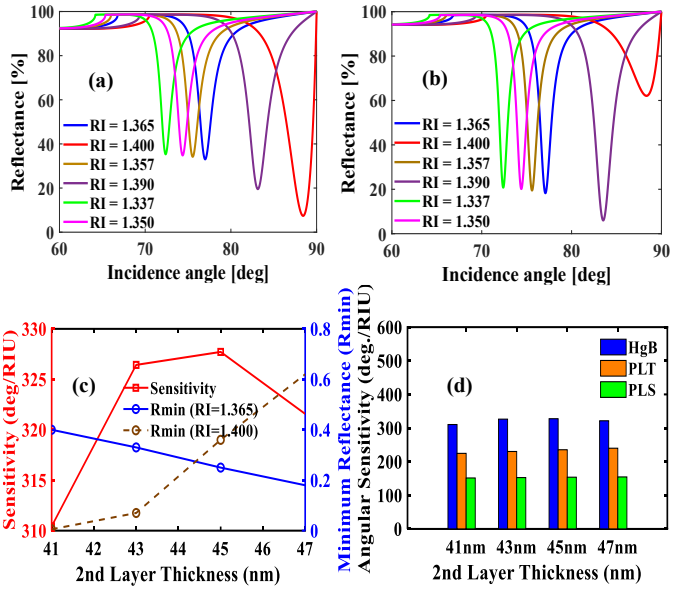


Fig. 5. SPR resonance curves for (a) d_{Cu} = 43nm and (b) d_{Cu} = 47nm. (c) Sensitivity (S) and minimum reflectance (R_{min}) vs. 2nd layer thickness. (d) Bar chart comparison of sensitivity for various thicknesses of the 2nd layer (41-47 nm) for the RI of 1.365 and 1.400 of the proposed biosensor.

TABLE IV.
ANGULAR SENSITIVITY FOR VARIOUS Cu LAYER THICKNESSES AT (d_{ZnSe} = 2 nm, d_{BP} = 0.53 nm)

Thickness of Cu Layer	Sensitivity of infected platelet (deg./RIU)	Sensitivity of infected plasma (deg./RIU)	Sensitivity of infected hemoglobin (deg./RIU)
41 nm	224.69	151.53	310.42
43 nm	230.30	152.69	326.42
45 nm	235.45	153.84	327.71
47 nm	240	154.61	321.57

As illustrated in Fig. 5(a) and Fig. 5(b), the reflectance spectra display a noticeable resonance dip that gradually shifts toward higher incident angles as the RI increases, confirming the RI-dependent response of the sensor. While a 43 nm layer produces a stronger resonant response, a slightly thicker 47 nm layer leads to broader and steeper dips. The relationship between sensitivity and minimum reflectance (R_{min}) for various Cu layer thicknesses 41 to 47 nm, is presented in Fig. 5(c). Sensitivity increases steadily up to 45 nm, reaching its peak and then slightly decreases, whereas R_{min} shows an opposite trend, suggesting that optimal plasmon coupling is at 45 nm. Finally, a bar chart comparing sensor performance across various thicknesses of the Cu layer is illustrated in Fig. 5(d), showing that 45 nm is the ideal thickness for the Cu layer of the biosensor.

D. Effect of variation in ZnSe Layer Thickness

Changing ZnSe layer thickness significantly influences the performance of SPR-based biosensors in detecting infected hemoglobin (Hgb), plasma (PLS), and platelet (PLT) cells. Table V exhibits that while increasing the ZnSe layer thickness from 1.5nm to 2.5 nm, the sensitivity of affected hemoglobin enhances, which reaches a maximum of 327.71 °/RIU at 2 nm, and then falls to 286.85 °/RIU at 2.5 nm.

TABLE V.
ANGULAR SENSITIVITY FOR VARIOUS ZnSe LAYER THICKNESSES AT (d_{Cu} = 45 nm, d_{BP} = 0.53 nm)

Thickness of ZnSe Layer	Sensitivity of infected platelet (deg./RIU)	Sensitivity of infected plasma (deg./RIU)	Sensitivity of infected hemoglobin (deg./RIU)
1.5 nm	220	149.61	291
2 nm	235.45	153.84	327.71
2.5 nm	256.36	158.46	286.85

In Fig. 6(a) and Fig.6(b), the reflectance versus incidence angle spectra for the thicknesses of 1.5 nm and 2.5 nm are plotted for RI ranging from 1.337 to 1.400. A gradual shift of the resonance angle toward higher values occurs as the RI increases, confirming the strong RI dependence and angular sensitivity of the sensor. The relationship between sensitivity and R_{min} with various ZnSe layer thicknesses of 1.5 to 2.5 nm is demonstrated in Fig.6(c). Sensitivity increases sharply with thickness up to 2 nm, then begins to decrease beyond this point. At the same time, the R_{min} curve gradually increases with increase in layer thickness, indicating weaker resonance coupling at larger thickness. A bar chart comparison of angular sensitivity for different ZnSe layer thicknesses is shown in Fig. 6(d), showing that a 2 nm ZnSe layer achieves the optimal sensor performance.

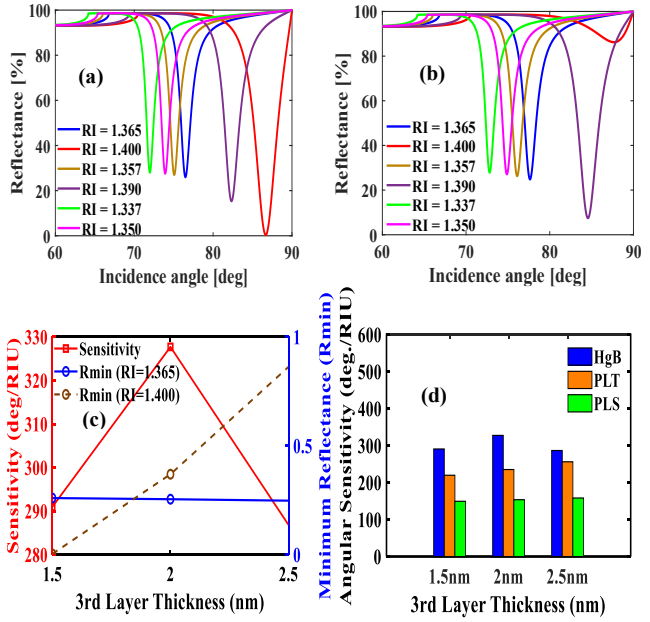


Fig. 6. SPR resonance curves for (a) d_{ZnSe} = 1.5nm and (b) d_{ZnSe} = 2.5nm. (c) Sensitivity (S) and minimum reflectance (R_{min}) vs. 3rd layer thickness. (d) Bar chart comparison of sensitivity for various thicknesses of the 3rd layer (1.5-2.5 nm) for the RI of 1.365 and 1.400 of the proposed biosensor.

E. Effect of varying the Number of 2D material layers

To improve overall performance, the number of 2D material (BP) layers is varied, and then we carefully analyze its impact on key performance parameters. The introduction of BP layers enhances the light–matter interaction at the metal-dielectric interface due to the high optical absorption and strong in-plane asymmetry of BP. This leads to stronger coupling between the incident p-polarized light and surface plasmon polaritons (SPPs). As observed from Fig. 7(a) and Fig. 7(b) the resonance angle shifts toward higher values with an increase in RI. Fig. 7(c) and Fig. 7(d) illustrate that the sensor achieves its maximum sensitivity of 327.71°/RIU when a single BP monolayer is used. However, this sensitivity gradually begins to decline as the number of layers increases during the detection of hemoglobin cells. On the other hand, the angular sensitivity of platelets and plasma improves when the number of BP layers rises, which is supported by the results summarized in Table VI. The RI clearly affects the R_{min} curve. At an RI of 1.365, the curve remains low and almost constant. However, when the RI rises to 1.400, the curve increases sharply with an increasing number of BP layers, indicating that the resonance quality declines at higher refractive indices.

TABLE VI.
ANGULAR SENSITIVITY FOR VARIOUS NUMBERS OF BP LAYERS AT (d_{Cu} = 45 nm, d_{ZnSe} = 2 nm)

Thickness of BP Layer	Sensitivity of infected platelet (deg./RIU)	Sensitivity of infected plasma (deg./RIU)	Sensitivity of infected hemoglobin (deg./RIU)
1x	235.45	153.84	327.71
2x	272.12	161.53	264.57
3x	335.75	170.38	189.74

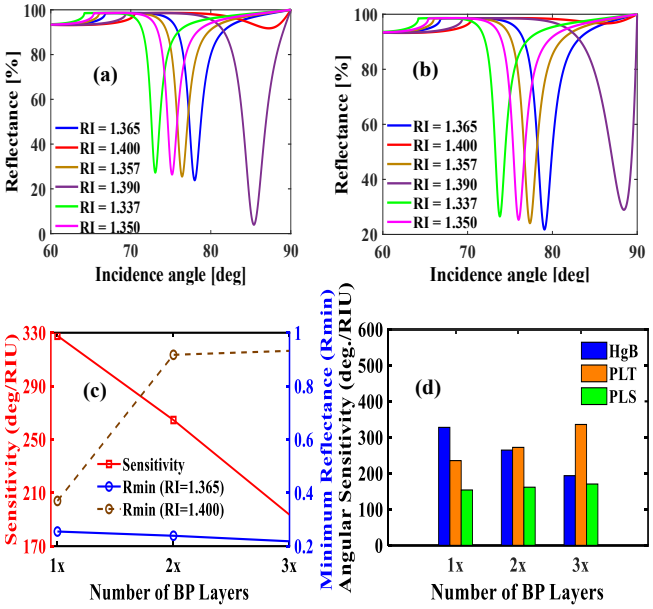


Fig. 7. SPR resonance curves for (a) two BP layers and (b) three BP layers. (c) Sensitivity (S) and minimum reflectance (R_{min}) vs. 4th layer thickness. (d) Bar chart comparison of sensitivity for various thicknesses of the 4th layer (0.53-1.59 nm) for the RI of 1.365 and 1.400 of the proposed biosensor.

F. Overall Sensor performance

Fig.8 shows how the reflection intensity profiles for six stages of dengue infection changes with the angle of incidence for various sensor setups.The design's performance improves

as different layers are deposited, as seen in Fig.8(a) through Fig.8(c).It illustrates how all of the individual layers contribute positively to the sensor's performance.

Fig.8(d) presents an analysis of QF and DA across various sensing media, showing that the N FK51A/Cu/ZnSe/BP/SM configuration exhibits high QF and exceptional DA, making it very effective for detection of dengue virus.

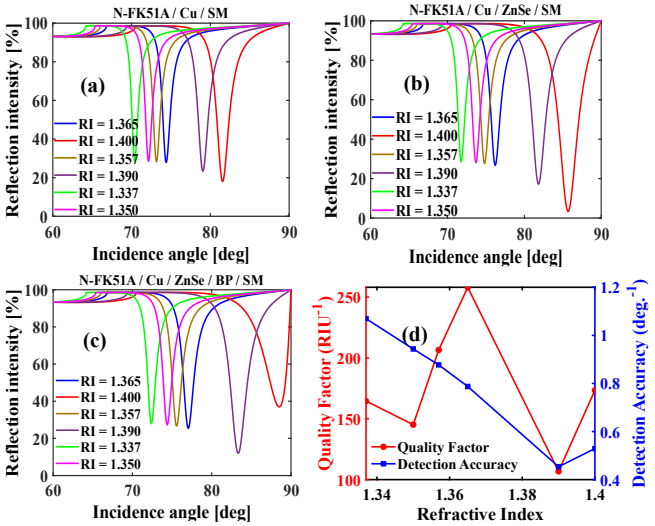


Fig. 8. Reflectance intensity curves for 6 different stages of dengue (a) N-FK51A / Cu /SM, (b) N-FK51A / Cu / ZnSe / SM, (c) N-FK51A / Cu / ZnSe /BP/ SM. (d) Comparison of quality factor and detection accuracy for different sensing media.

As shown in Fig. 9, the reflectance behavior varies with the angle of incidence for various blood components, such as Hgb, PLT, and PLS under both normal (N) and affected (A) conditions. The simulation results obtained from both COMSOL Multiphysics and MATLAB show a strong agreement, revealing uniform shifts in resonant angle across the refractive index range of 1.337-1.400. This confirms the computational accuracy of both methods. Finally, we compared the proposed biosensor with previously reported designs as indicated in Table VII. Under comparable configurations, the recommended structure functions exceptionally well.

TABLE VII.
COMPARISON WITH RECENT STUDIES

Ref.	Year	Structure	Sensitivity (deg./RIU)	FWHM (deg.)	QF (RIU^{-1})
This Work	2025	NFK51A/Cu/ZnSe /BP	327.71	1.89	156.68
[5]	2025	BK7/Ag/ TiO_2/BP	309.74	3.53	87.54
[13]	2025	BK7/TiO2/Ag/TiO_2/WSe2	288.11	2.66	108.27
[7]	2024	BK7/Ag/AlN/BP	310.29	2.01	147.31

IV. CONCLUSION

A multilayered SPR biosensor developed with the N-FK51A/Cu/ZnSe/BP configuration has been designed to detect the dengue virus and evaluated by numerical modeling. The sensing medium comprises platelets, plasma, and hemoglobin cells from healthy and dengue-infected blood samples. The combination of ZnSe and BP materials provides the sensor with sharper resonance curves, improved detection accuracy, and better field confinement than conventional metal-based sensors. Systematic optimization of the sensor produced peak performance with an angular sensitivity of 327.71°/RIU, a QF of 156.68 RIU^{-1}, and a high detection accuracy of 0.78745 deg^{-1}. The optimized material and layer design greatly enhance the plasmonic response, making the sensor an efficient, label-free, and low-cost diagnostic tool for clinical and biomedical use.

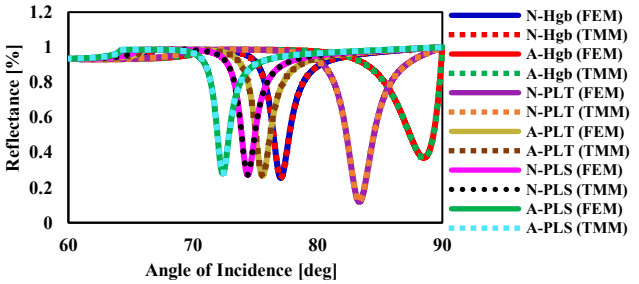


Fig. 9. Reflectance intensity curves with the change of incident angle of the proposed sensor derived from COMSOL and MATLAB for blood sample's RI of 1.337 to 1.400

The novelty of this work does not lie in using SPR alone for dengue detection, but in the specific material engineering, optimization strategy, and dual-method validation that together produce a measurably superior sensing performance. This work presents a new multilayer combination for the detection of the dengue virus. This precise combination of N-FK51A / Cu / ZnSe /BP/ SM has not been previously reported. This low-cost material stack can outperform several Ag/Au-based sensors in several performance parameters.

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